

Ethan Sotka

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EDUCATION

University of Notre Dame | Notre Dame, IN

Dec 2026

Bachelor of Science: Computer Science, Applied Computational Mathematics and Statistics

GPA: 3.9

Relevant Coursework: Linear Algebra, Probability, Statistical Methods, Differential Equations, Data Structures

TECHNICAL AND LANGUAGE SKILLS

Technical: Python(*Advanced*), Excel(*Advanced*), C(*Intermediate*), C++(*Intermediate*), MATLAB(*Intermediate*), R(*Intermediate*), Git(*Basic*), Java(*Basic*)

Frameworks: NumPy(*Intermediate*), Pandas(*Intermediate*), Matplotlib/Seaborn(*Intermediate*), BeautifulSoup(*Basic*), TensorFlow(*Basic*)

WORK EXPERIENCE

Exiger | McLean, VA

Starting May 2025

Data Science Intern

- Leverage **Python** to analyze supply chain data, identify weak links, and propose data-driven solutions for operational resilience

System 0 Thinking - Can AI Think | *Data Inference and Analytics and Learning Lab*

January 2025 - May 2025

- Developed **Python** based program leveraging **ChatGPT API** with knowledge graph as a **RAG** to test if LLMs held core beliefs

Cornerstone Remodeling | Columbia, MD

May 2023 – August 2023

Assistant to Craftsman

- Collaborated with craftsmen on house remodeling and expansion projects of \$400,000+, showcasing teamwork, communication, and adaptability while independently completing demolition tasks with a focus on safety, precision, and construction best practices
- Developed proficiency with a diverse range of tools, emphasizing safety and effectiveness to ensure high-quality workmanship.

ACTIVITIES AND LEADERSHIP

Quant Club of Notre Dame | Notre Dame, IN

August 2023 – Present

Officer

- Applied Game Theory, Probability, and Risk Management in real-time simulations through strategy games and competitions
- Explored quantitative finance best practices through lectures involving professional Quants to understand Quantitative trading

DELTA (Discovering Econometrics: Learning Through Application) | Notre Dame, IN

August 2024 – Present

Member

- Applied mathematical and statistical methods to solve real-world problems across various disciplines in short time frames
- Developed an advanced understanding of quantitative tools used by economists and data scientists through professional oversight

DataFest 2024 | Notre Dame, IN

Finalist

March 2024

- Analyzed several data sets of 10,000+ points with **numpy** and **pandas** for CourseKata to improve user learning and enjoyment
- Presented trends and patterns in data to CourseKata along with actionable next steps to allow advancement of teaching practices

Investment Club | Glenelg, MD

August 2022 – May 2023

Co-Founder and Co-Leader

- Secured \$5000 to invest as a club by pitching idea of club to our network and donated money to the school at the end of the year
- Taught members investment strategies by expert lectures in club meetings and an investment competition with paper accounts

PROJECTS

Predicting Longevity/Success at QB in the NFL | Stat Methods & Data Analysis August 2024 - Dec 2024

- Applied multiple Linear Regression with tests for assumptions to evaluate college football statistics using **R**, to develop predictive models for projecting NFL quarterback win rates and discovering no predictive power for longevity in the NFL
- Submitted research paper to **Baltimore Ravens**, with positive feedback from Senior Manager and invitation for further research

GDP Correlation Analysis App | Engineering Computing

January 2024 - May 2024

- Analyzed GDP growth of over 200 countries in an 80-year period with a multitude of factors to find correlation of GDP growth
- Utilized linear regression, heat maps, and graphic data in **MATLAB's** app feature to display the variables that affect GDP growth

Sports Betting Arbitrage Tool | Quant Club

March 2024 - April 2024

- Utilized **Selenium**, **BeautifulSoup**, and **Pandas** in **Python** to repeatedly scrape betting odds from several sports betting websites
- Applied Positive Expected Value and Arbitrage strategies with Kelly Criterion to data in simulation to create positive returns

Interests: Football, Entrepreneurship, Neuroscience, Fitness, Coding, Jiu Jitsu, Investing, Baltimore Ravens, Strategic/Number Games